

CMSC 10 – Introduction to Computer Science

Generation and Components of Computers

Week 2/16

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Generation of Computers

First Generation (1940s-1950s):

Key Technology: Vacuum tubes

Notable Computers: ENIAC, UNIVAC I, EDVAC

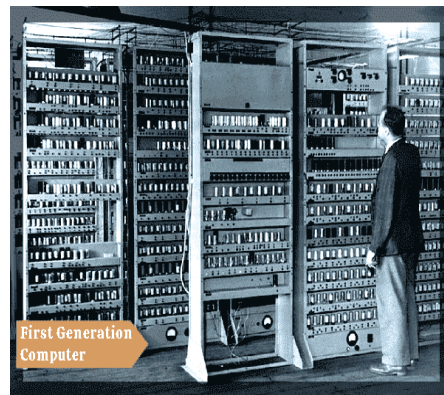
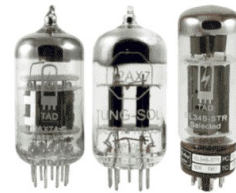
Characteristics:

- Large, room-sized machines.
- High power consumption and heat generation.
- Relatively slow processing speeds.
- Limited memory and storage capacity.

Physical Size: ENIAC occupied a significant amount of space. It consisted of 30 separate cabinets x 2.7 m tall x 0.79 m wide x 0.61 m deep.

Weight = 30 tons

Total area used = 167 square meters.



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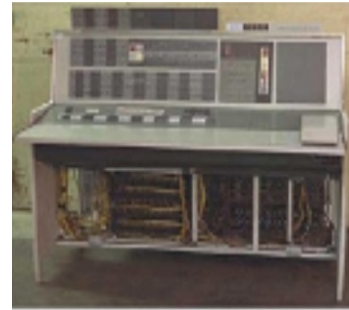
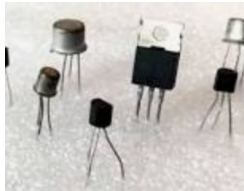
Generation of Computers

Second Generation (1950s-1960s):

Key Technology: Transistors

Notable Computers: IBM 700 series, UNIVAC III
DEC PDP-1

Characteristics: Smaller, more reliable, faster, and required less power than first-generation computers.



Second Generation



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Generation of Computers

Third Generation (1960s-1970s):

Key Technology: Integrated Circuits (ICs)

Notable Computers: IBM System/360,
DEC PDP-11, CDC 6600

Characteristics:

- Much smaller and more powerful than previous generations,
- allowed for greater miniaturization, and
- facilitated the development of smaller mainframes and minicomputers.



Third Generation

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Generation of Computers

Fourth Generation (1970s-1980s):

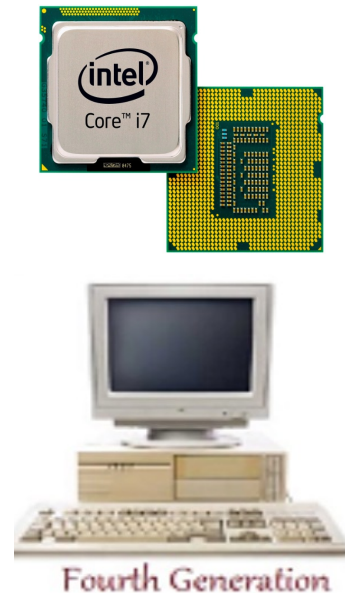
Key Technology: Microprocessors

Notable Computers:

- IBM 5150 (IBM PC),
- Apple II, Commodore 64

Characteristics:

- Introduction of microprocessors led to personal computers (PCs), which were much smaller and affordable for individuals and businesses.



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Generation of Computers

Fifth Generation (1980s-Present):

Key Technology:

- VLSI (Very Large Scale Integration) microprocessors,
- parallel processing, AI

Notable Computers:

- Various personal computers, laptops, smartphones

Characteristics:

- Ongoing advancements in microprocessor technology,
- the rise of the internet, and the
- development of artificial intelligence (AI) have defined this generation.
- Computers are smaller, more powerful, and interconnected on a global scale



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Computers Then

- Computers were fairly mysterious devices that only a few people knew about.
- Only companies/big organizations can afford to buy them (P5,000,000)
- Only those people in these organizations can use them
- You need a big space for only 1 computer



Price of Computer = P5M
 Space required = 10 sq m
 (equivalent to 1 lot of low cost subdivision)

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Computers Today



Price of Computer = P25K-P50K
 Space required = 1 sq ft

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Categories of Computers

Computers can be classified into various categories based on their size, processing power, purpose, and architecture.

1. Supercomputers

- the most powerful and fastest computers available
- Designed for performing complex scientific and engineering calculations, simulations, and data-intensive tasks.
- Applications: weather forecasting, nuclear research, and molecular modeling, quantum mechanics, weather forecasting, oil and gas exploration, molecular modeling, physical simulations, aerodynamics, nuclear fusion research and cryptanalysis.
- Examples:
 - Frontier (1,102 PFLOPS).
 - Fugaku (442 PFLOPS)
 - LUMI (309 PFLOPS)
 - Leonardo (174 PFLOPS)
 - Summit (148 PFLOPS)
 - Sierra (94 PFLOPS)
 - SunwayTaihuLight (93 PFLOPS)
 - Perlmutter (70 PFLOPS)

A petaflop (PFLOP is a unit of computing speed equal to one quadrillion floating-point operations per second.

1,102 PFLOPS = 1.102×10^{15} floating-point operations per second, making it an extremely powerful computing system.

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Categories of Computers

2. Mainframe Computers

- Mainframes are large, high-performance computers that can handle multiple users and process massive amounts of data simultaneously.
- used by large organizations for critical applications like banking, airline reservations, and database management.
- Mainframes are data servers designed to process up to 1 trillion web transactions daily with the highest levels of security and reliability.
- Mainframes are high-performance computers with large amounts of memory and data processors that process billions of simple calculations and transactions in real time.
- A mainframe computer is critical to commercial databases, transaction servers, and applications that require high resiliency, security and agility.

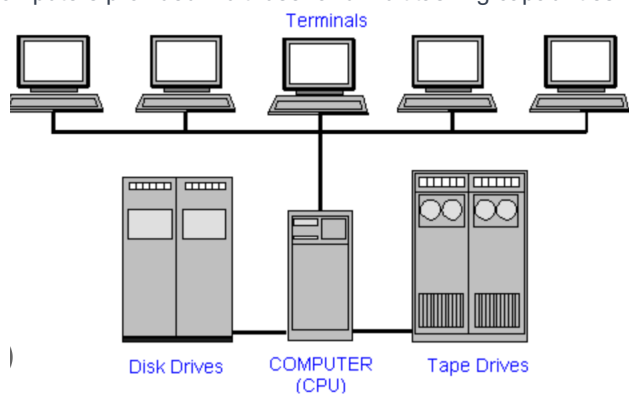


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Categories of Computers

3. Minicomputers

- smaller than mainframes but more powerful than personal computers.
- They were widely used in businesses and research institutions before the rise of PCs.
- Minicomputers provided multi-user and multitasking capabilities.



<https://www.pcmag.com/encyclopedia/term/minicomputer>

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Categories of Computers

4. Workstations

- Workstations are high-performance computers designed for specialized tasks, such as graphic design, 3D modeling, scientific simulations, and software development.
- They offer advanced graphics and processing capabilities.



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Categories of Computers

5. Personal Computers (PCs)

- Personal computers are designed for individual use and come in various forms, including desktops, laptops, and tablets.
- They are used for general-purpose computing, such as word processing, web browsing, gaming, and entertainment.

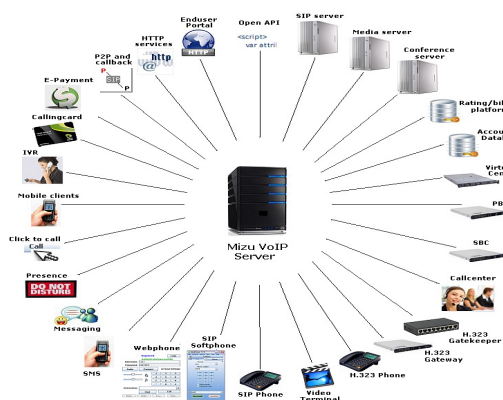


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Categories of Computers

6. Servers

- Servers are computers optimized for providing services, resources, or data to other computers or devices over a network.
- They come in various types, including web servers, file servers, and database servers, and are essential for supporting networked applications.



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Categories of Computers

7. Wearable Computers

- Wearable computers are small, lightweight devices that can be worn on the body.
- Examples include smartwatches and fitness trackers, which provide features like health monitoring and notifications.

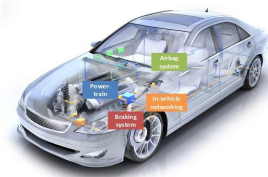


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Categories of Computers

8. Embedded Computers

- Embedded computers are specialized computers integrated into other devices or systems, such as appliances, automobiles, industrial machines, and consumer electronics. They perform dedicated functions and are often designed for low-power and real-time operations.



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Categories of Computers

9. Smartphones and Tablets

- These are compact, portable devices that combine computing power with communication capabilities.
- They are used for a wide range of tasks, including communication, web browsing, app usage, and entertainment.



10. Cloud Computing Infrastructure

- Cloud computing involves a network of remote servers that provide various computing resources and services over the internet.
- Users can access computing power, storage, and software on-demand without owning physical hardware.

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Thank you!

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